Exercises for Section 9.1

Directions: For exercises 1 - 5, determine whether the given ordered pair is a solution to the system of equations.

1. $\begin{aligned} 5x - y &= 4 \\ x + 6y &= 2 \end{aligned}$ and $(4,0)$

2. $\begin{aligned} -3x - 5y &= 13 \\ -x + 4y &= 10 \end{aligned}$ and $(-6,1)$

3. $\begin{aligned} 3x + 7y &= 1 \\ 2x + 4y &= 0 \end{aligned}$ and $(2,3)$

4. $\begin{aligned} -2x + 5y &= 7 \\ 2x + 9y &= 7 \end{aligned}$ and $(-1,1)$

5. $\begin{aligned} x + 8y &= 43 \\ 3x - 2y &= -1 \end{aligned}$ and $(3,5)$

Directions: For exercises 6 - 15, solve each system by substitution.

6. $\begin{aligned} x + 3y &= 5 \\ 2x + 3y &= 4 \end{aligned}$

7. $\begin{aligned} 3x - 2y &= 18 \\ 5x + 10y &= -10 \end{aligned}$

8. $\begin{aligned} 4x + 2y &= -10 \\ 3x + 9y &= 0 \end{aligned}$

9. $\begin{aligned} 2x + 4y &= -3.8 \\ 9x - 5y &= 1.3 \end{aligned}$

10. $\begin{aligned} -2x + 3y &= 1.2 \\ -3x - 6y &= 1.8 \end{aligned}$

11. $\begin{aligned} x - 0.2y &= 1 \\ -10x + 2y &= 5 \end{aligned}$

12. $\begin{aligned} 3x + 5y &= 9 \\ 30x + 50y &= -90 \end{aligned}$

13. $\begin{aligned} -3x + y &= 2 \\ 12x - 4y &= -8 \end{aligned}$

14. $\begin{aligned} \tfrac{1}{2}x + \tfrac{1}{3}y &= 16 \\ \tfrac{1}{6}x + \tfrac{1}{4}y &= 9 \end{aligned}$

15. $\begin{aligned} -\tfrac{1}{4}x + \tfrac{3}{2}y &= 11 \\ -\tfrac{1}{8}x + \tfrac{1}{3}y &= 3 \end{aligned}$

Directions: For exercises 16 - 25, solve each system by addition.

16. $\begin{array}{l} -2x+5y=-42 \\ 7x+2y=30 \end{array}$

17. $\begin{array}{l} 6x-5y=-34 \\ 2x+6y=4 \end{array}$

18. $\begin{array}{l} 5x-y=-2.6 \\ -4x-6y=1.4 \end{array}$

19. $\begin{array}{l} 7x-2y=3 \\ 4x+5y=3.25 \end{array}$

20. $\begin{array}{l} -\mathrm{x}+2y=-1 \\ 5x-10y=6 \end{array}$

21. $\begin{array}{l} 7x+6y=2 \\ -28x-24y=-8 \end{array}$

22. $\begin{array}{l} \frac{5}{6}x+\frac{1}{4}y=0 \\ \frac{1}{8}x-\frac{1}{2}y=-\frac{43}{120} \end{array}$

23. $\begin{array}{l} \frac{1}{3}x+\frac{1}{9}y=\frac{2}{9} \\ -\frac{1}{2}x+\frac{4}{5}y=-\frac{1}{3} \end{array}$

24. $\begin{array}{l} -0.2x+0.4y=0.6 \\ x-2y=-3 \end{array}$

25. $\begin{array}{l} -0.1x+0.2y=0.6 \\ 5x-10y=1 \end{array}$

Directions: For exercises 26 - 35, solve each system by any method.

26. $\begin{array}{l} 5x+9y=16 \\ x+2y=4 \end{array}$

27. $\begin{array}{l} 6x-8y=-0.6 \\ 3x+2y=0.9 \end{array}$

28. $\begin{array}{l} 5x-2y=2.25 \\ 7x-4y=3 \end{array}$

29. $\begin{array}{l} x-\frac{5}{12}y=-\frac{55}{12} \\ -6x+\frac{5}{2}y=\frac{55}{2} \end{array}$

30. $\begin{array}{l} 7x-4y=\frac{7}{6} \\ 2x+4y=\frac{1}{3} \end{array}$

31. $\begin{array}{l} 3x + 6y = 11 \\ 2x + 4y = 9 \end{array}$

32. $\begin{array}{l} \frac{7}{3}x - \frac{1}{6}y = 2 \\ -\frac{21}{6}x + \frac{3}{12}y = -3 \end{array}$

33. $\begin{array}{l} \frac{1}{2}x + \frac{1}{3}y = \frac{1}{3} \\ \frac{3}{2}x + \frac{1}{4}y = -\frac{1}{8} \end{array}$

34. $\begin{array}{l} 2.2x + 1.3y = -0.1 \\ 4.2x + 4.2y = 2.1 \end{array}$

35. $\begin{array}{l} 0.1x + 0.2y = 2 \\ 0.35x - 0.3y = 0 \end{array}$

Directions: For exercises 36 - 40, graph the system of equations and state whether the system is consistent or inconsistent, and whether the system has no solution, one solution, or infinitely many solutions.

36. $\begin{array}{l} 3x - y = 0.6 \\ x - 2y = 1.3 \end{array}$

37. $\begin{array}{l} -x + 2y = 4 \\ 2x - 4y = 1 \end{array}$

38. $\begin{array}{l} x + 2y = 7 \\ 2x + 6y = 12 \end{array}$

39. $\begin{array}{l} 3x - 5y = 7 \\ x - 2y = 3 \end{array}$

40. $\begin{array}{l} 3x - 2y = 5 \\ -9x + 6y = -15 \end{array}$